

Debugging and Testing Web Applications in a Linux Environment

Learning Objectives

By the end of this lab, students will be able to:

- Understand the difference between testing and debugging.
- Use basic Linux commands to navigate and manage files.
- Run and debug a simple Python web application.
- Use tools like Flask, logs and Vim to identify and fix bugs.
- Write and run basic unit tests using pytest.

Lab Setup

This lab uses a Coursera Linux VM with the following tools pre-installed:

- Python 3
- pytest
- A sample buggy web app (provided in the lab folder)

Step-by-Step Instructions

Step 1: Navigate to the Lab Folder

Open the terminal and run:

```
cd /home/coder/Desktop/exercise  
ls
```

You should see files like:

- app.py or test_app.py
- README.md

```
python3 -m pip install --user flask
```

Step 2: Run the Application

```
python3 app.py
```

When you run “python3 app.py “ you will get this error: “ImportError: cannot import name 'Flsk' from 'flask’”

That’s expected, because the code has Flsk instead of Flask.

Use curl to test endpoints:

```
curl http://localhost:5000/
```

Step 3: Identify and Fix Bugs

1. Read the error messages in the terminal.
2. Open the code using
3. vim app.py
4. Look for syntax errors, logic errors, or missing imports
5. Edit the code and re-run the application until it works correctly.
6. Press i → to enter insert mode
7. Make your edits (change Flsk → Flask, fix the other bugs)
8. Press Esc → then type :wq → Enter to save & quit

Run the app: python3 app.py

You should see : * Running on http://127.0.0.1:5000

Step 4: Run Unit Tests

1. Test manually with curl
In another terminal (don’t stop the server yet):
curl http://localhost:5000/
curl <http://localhost:5000/add>
Expected results:
{“message”:“Hello world”}
{“result”:4}
2. Stop the server: Go back to the first terminal where the server is running and press CTRL+C.
3. Run automated tests. Make sure pytest is installed:

```
python3 -m pip install --user pytest
```

```
python3 -m pytest -q
pytest-3 test_app.py
```

Fix any failing tests and re-run them until all pass.

Expected:

```
..
```

```
2 passed in 0.xx s
```

Step 5: Reflect

Create a short report answering the following:

- What bugs did you find?
- How did you fix them?
- What did the tests reveal?